

Lappeenranta teknillinen yliopisto
School of Business and Management

Software Development Skills

André Olkinuora

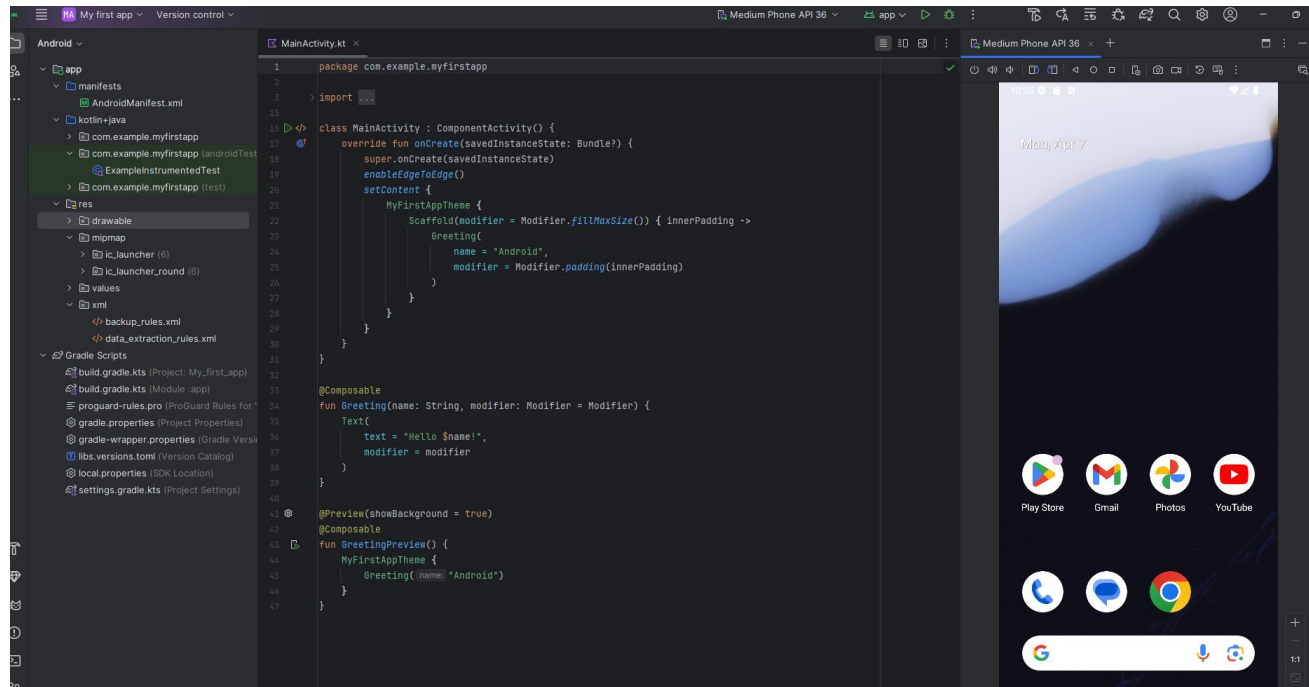
LEARNING DIARY, Software Development Skills: Mobile

NOTE:

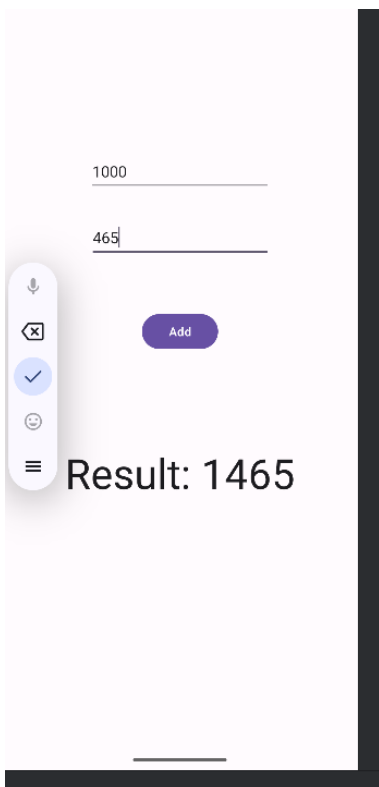
All tasks and the project are in the same GitHub repository, each within its own branch.

LEARNING DIARY

7.4.2025



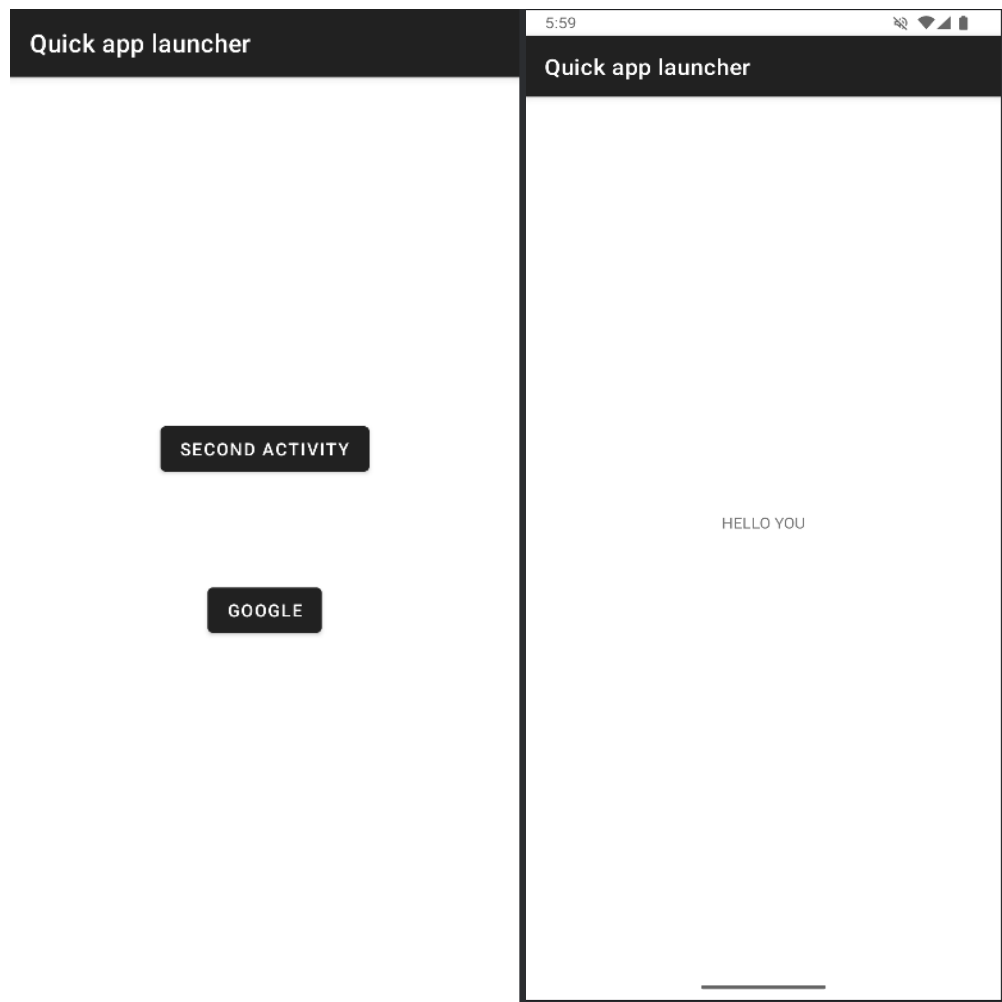
First time opening android studio



I started with going through the introduction lesson. First thing I had to do, is actually do some research on why everything is so different from the video tutorial you are supposed to follow. So, after a while I figured out how to navigate the program and find things that are nowadays different. For example, to get the ability to “use the pallet” instead of “writing the textboxes in code”, I had to choose different option when choosing the project type etc. So instead of “Empty Activity”, choosing “Empty Views Activity” allows to use the old style of creating apps. The introduction lesson was fairly simple and clearly was just used to familiarize the students with the program.

8.4.2025

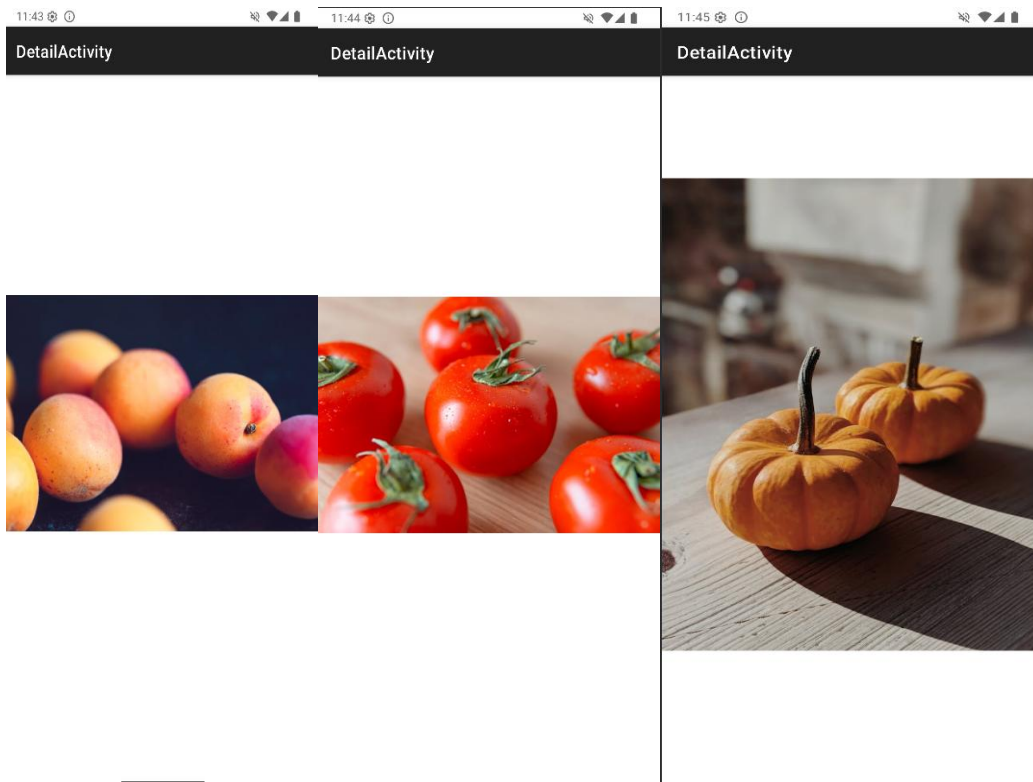
I started to work on core-elements “lesson”. It felt fairly simple and no big problems came up except for the fact that the video really is pretty outdated so for example the `resolveActivity` can’t be used anymore, but simple `startActivity` will work. It really has surprised me how simple app creation is after all, or at least I expected it to be a lot harder since this is my first mobile app course that I have taken. Still I feel like these lessons have been extremely helpful even though the tutorial is a bit outdated, but at least it forces to think bit more since you can’t straight up copy the tutorial. Another thing I’ve noticed that how much more I prefer using the VSC editor than android studios own, but since in this case it’s million times easier to keep using android studio, I’ll just have to bear with it.



9.4.2025

Continued to the next “lesson”. It’s pretty common for there to be a “leap” on how challenging tasks are in every course I have taken and I feel like this was the leap on this course, or if not challenging, at least takes more time to complete. Felt like this one took a lot longer to finish and was a bit more challenging, not much, but still clear difference in my opinion. Of course the biggest difference maker was the challenge in “translating / doing the same thing” in kt as the video tutorial does. Still, managed to finish the lesson without too many issues.

List app	
peach	\$0.99
Fresh peaches	
tomato	\$1.49
Fresh salad	
squash	\$0.89
Fresh yellow squash	



14.4.2025

Started the project work of the course. Had hard time coming up with a decent idea, but decided to go with making a "Habit reminder app", which basically allows the user to add habits (or just use it as normal reminder app) and set a reminder time for it and the app will send a notification of it every day at that time.

I started with adding the logic for simply adding new habits, a checkbox for them and deleting them. That part was fairly simple and was basically just doing the same as in previous "lessons". Next up, I added the functionality for editing already added habits, which also was fairly simple.

The thing I mostly had issues with was actually figuring out how to save the states of the habits but quickly found out resolutions for that too.

15.4.2025

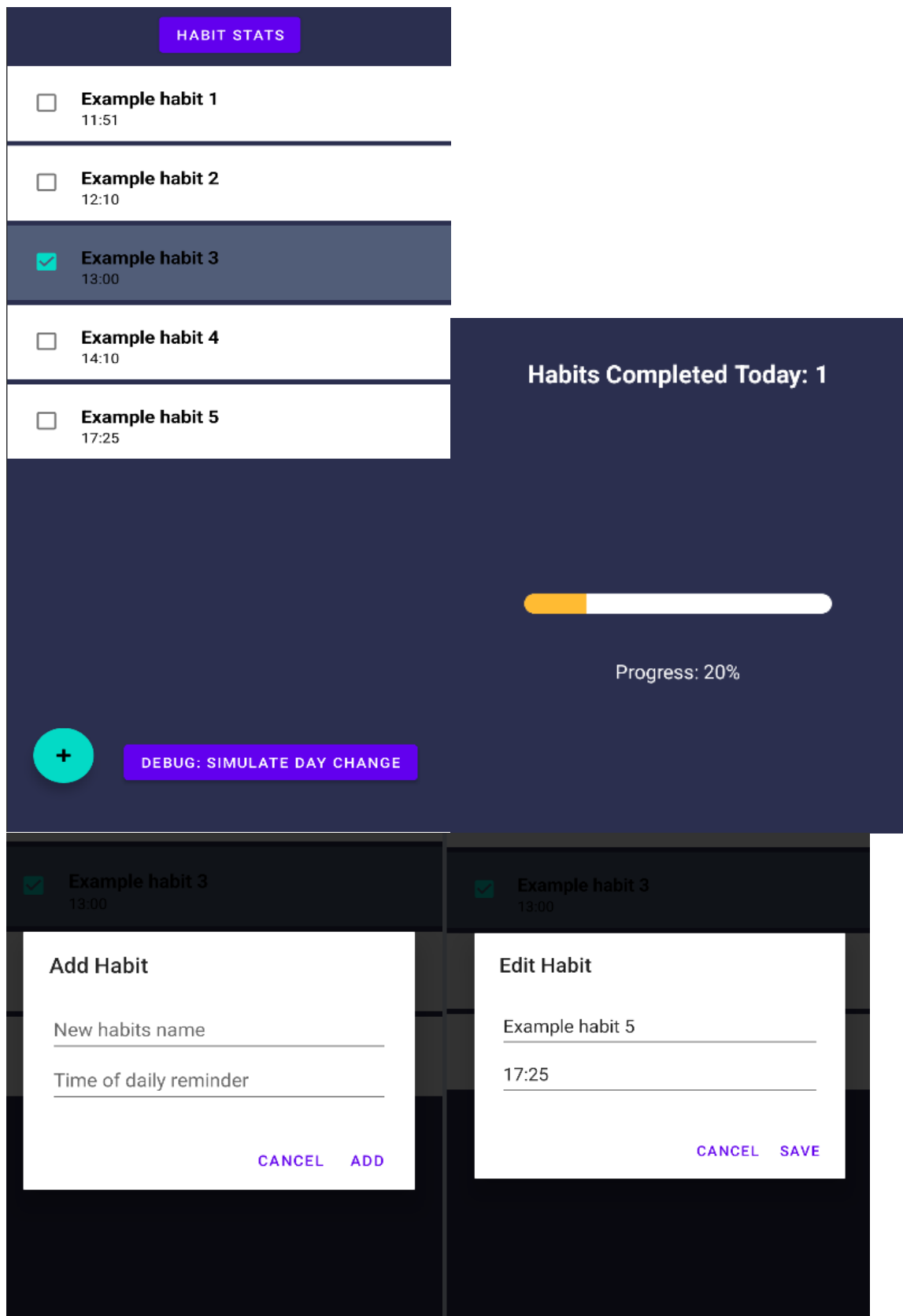
Continued the project work, since my app was starting to shape up well, I decided to dive into the notification part of the app, and that was actually the most problematic part of the development. I managed to get the notifications to SOMETIMES pop-up, but it was not consistent at all, I added debugging buttons etc. and could not figure out why the notifications wouldn't pop up every time when I set them with the timer functionality. Finally, after some digging I found it:

```
<uses-permission android:name="android.permission.SCHEDULE_EXACT_ALARM" />
```

This permission is what finally allowed the timer to function consistently, what I still don't quite understand is why it did work every now and then? Like, you would think the timed notifications wouldn't go through at all, without that, but for some reason they did. Otherwise, this part went through well too, so there were no bigger problems after this. One thing I noticed that the app still does needs to ask for permission to "set alarms" etc. but that is understandable.

Since the user checks the checkboxes every time, they complete the habit, it also needs to be reset since the habits are supposed to be daily. I decided to make it so that at midnight, the checkboxes get unchecked. I also added a debug button to simulate this so that I can easily show it functioning in the video I need to record for the course. I also added a stats page, where the user can check out how many habits they have completed and the percentage of them for every day.

All in all, the project part was fairly simple too, but so was my program which of course affected it, biggest problems I faced were all somehow connected to permissions etc. And there were a lot of places to find solutions on them too, even if they have changed a lot with different Android versions over time.



NOTE:

All tasks and the project are in the same GitHub repository, each within its own branch.